

InViTAG – a web application for AI-assisted exploration and grouping of scientific images and data

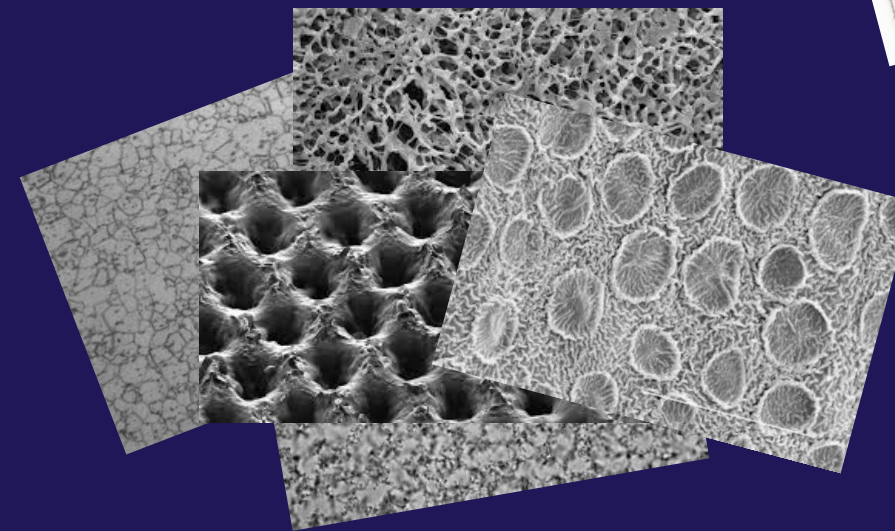
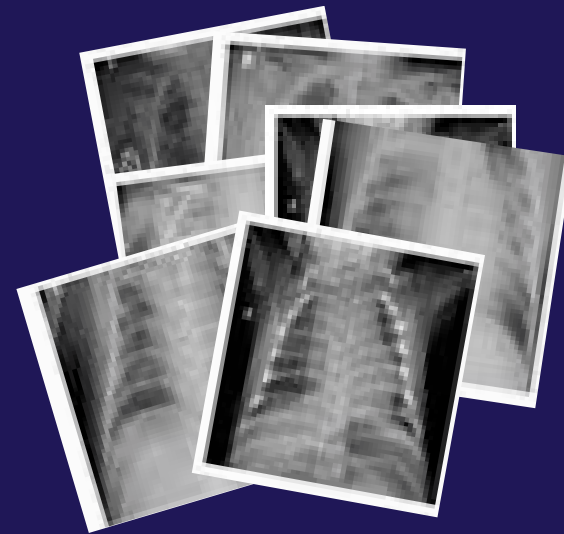
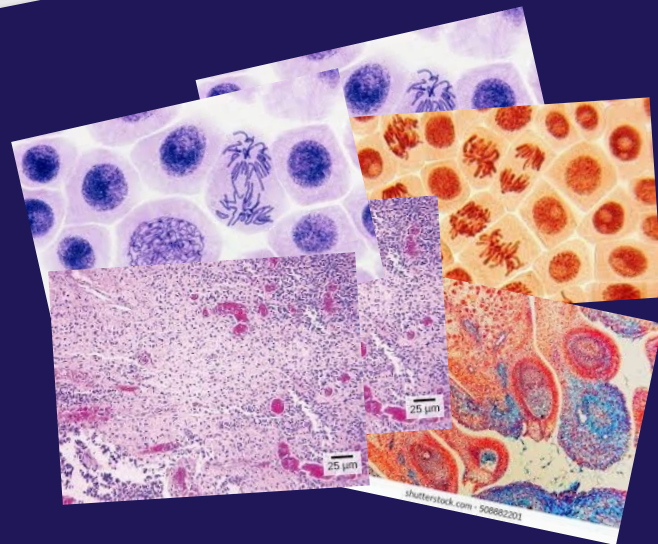
Ala Abuthawabeh, Amman Arab University

Michaël Aupetit, Qatar Computing Research Institute, HBKU

QEERI – MAY 13, 2026

For data-scientists or domain experts

Exploring scientific image data



For data-scientists or domain experts

Exploring scientific image data

*Accelerometer data from wearable devices
worn for a full week by 264 patients
with different overweight / obesity conditions*

device_id	segment_id	start_time	end_time	activity_level
712	1	2/8/2015 14:00	2/8/2015 14:50	moderate
712	1	2/8/2015 14:51	2/8/2015 15:37	light
712	1	2/8/2015 15:38	2/8/2015 15:39	sedentary
712	1	2/8/2015 15:40	2/8/2015 18:20	sleep
712	2	2/8/2015 18:21	2/8/2015 18:50	light
712	2	2/8/2015 18:51	2/8/2015 19:20	moderate
712	2	2/8/2015 19:21	2/8/2015 19:52	moderate
712	2	2/8/2015 19:53	2/8/2015 20:52	sleep
712	3	2/10/2015 22:13	2/10/2015 22:50	light
712	3	2/10/2015 22:51	2/10/2015 23:04	sedentary
712	3	2/10/2015 23:05	2/11/2015 22:11	sleep

For data-scientists or domain experts

Exploring scientific image data

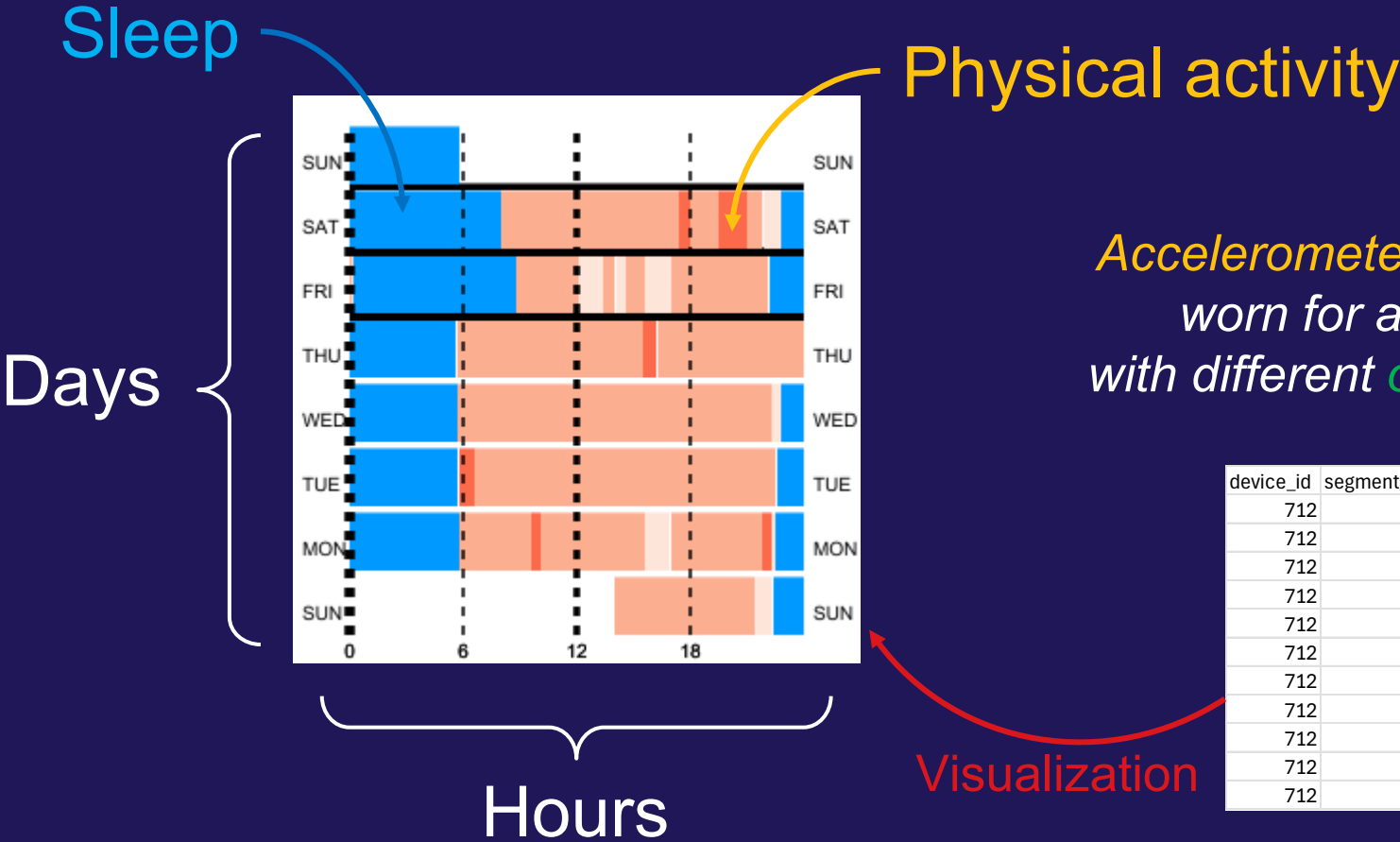
Accelerometer data from wearable devices
worn for a *full week* by *264 patients*
with different *overweight / obesity* conditions

device_id	segment	start_time	end_time	activity_level
712	1	2/8/2015 14:00	2/8/2015 14:50	moderate
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712	3	2/10/2015 22:51	2/10/2015 23:04	sedentary
712	3	2/10/2015 23:05	2/11/2015 22:11	sleep

Features								
image_id	group_id	group_color	wd_avg_ls_sleep_mid_time	wd_avg_ls_sleep_duration	wd_avg_sleep_duration	wd_num_sleep_segts_only	wd_num_all_sleep_segts	wd_num_sedentary_segts
695	group4	#66C2A5	14.14	8.71	5.2	13	2	14
696	outlier	#E78AC3	10.45	17.05	17.05	1	0	1
697	Uncategor	#D3D3D3	15.27	6.23	4.42	11	1	16
698	Uncategor	#D3D3D3	13.6	7.85	5.22	9	2	26
699	Uncategor	#D3D3D3	15.22	9.22	6.8	10	0	21

For data-scientists or domain experts

Exploring scientific image data



Accelerometer data from wearable devices worn for a full week by 264 patients with different overweight / obesity conditions

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712	1	2/8/2015 14:51	2/8/2015 15:37	light
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712	3	2/10/2015 22:51	2/10/2015 23:04	sedentary
712	3	2/10/2015 23:05	2/11/2015 22:11	sleep

Visualization

Features

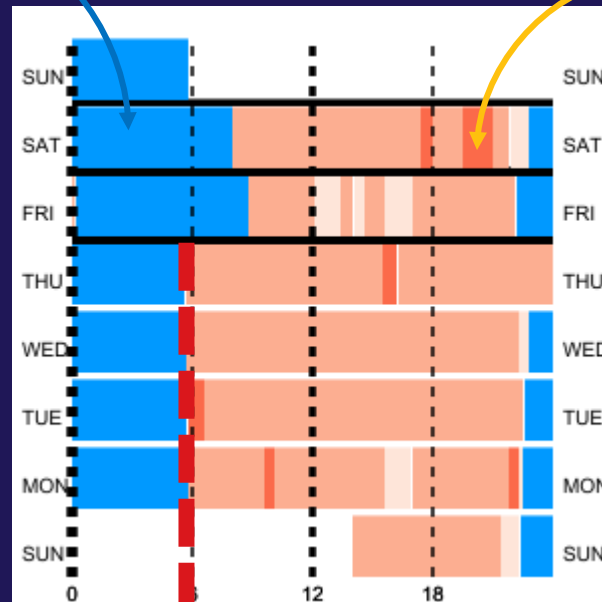
image_id	group_id	group_color	wd_avg_sleep_mid_time	wd_avg_sleep_duration	wd_avg_sleep_duration	wd_num_sleep_segts_only	wd_num_all_sleep_segts	wd_num_sedentary_segts
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699	Uncategor	#D3D3D3	15.22	9.22	6.8	10	0	21

For data-scientists or domain experts

Exploring scientific image data

Sleep

Physical activity

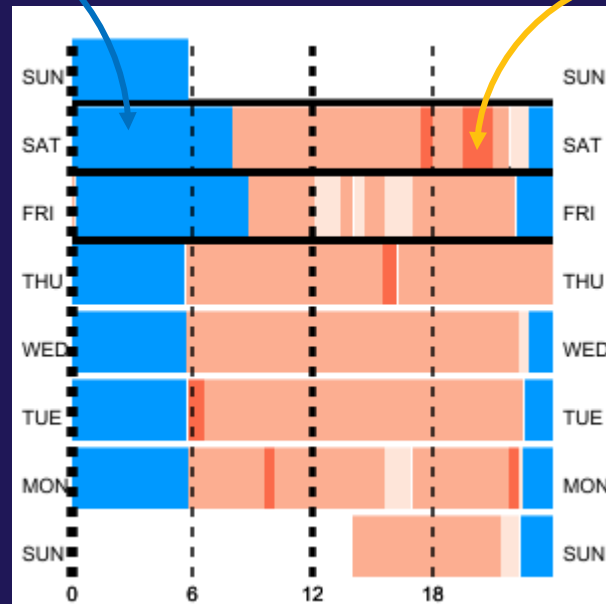


Regular
wake-up
time

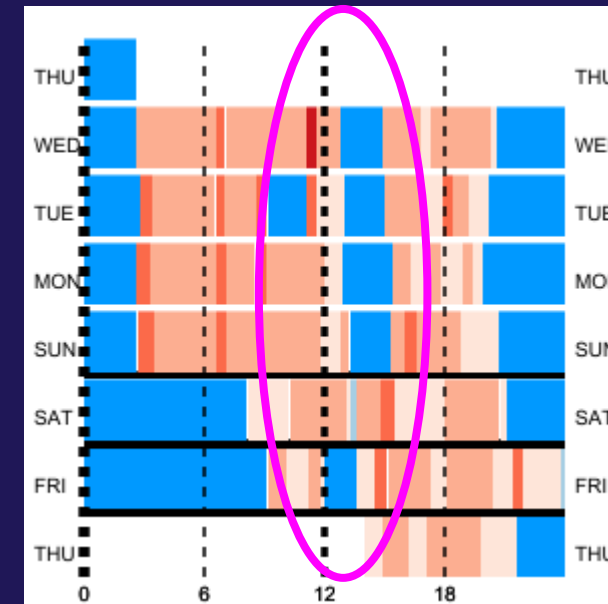
For data-scientists or domain experts

Exploring scientific image data

Sleep



Physical activity

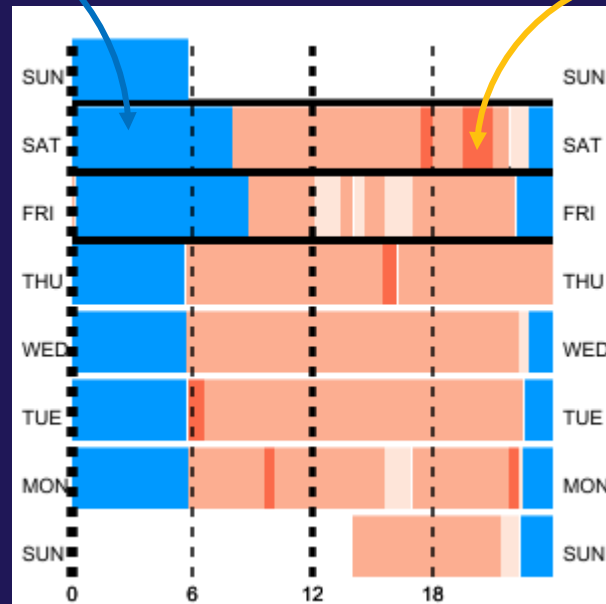


Naps

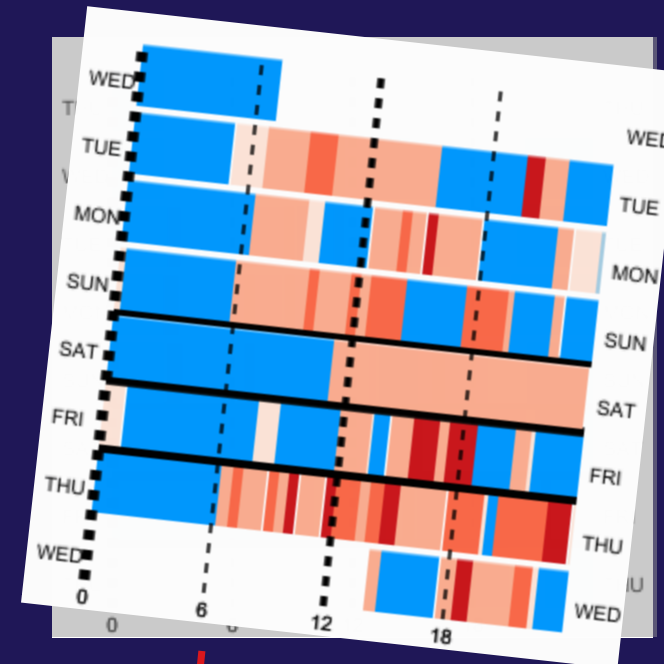
For data-scientists or domain experts

Exploring scientific image data

Sleep



Physical activity

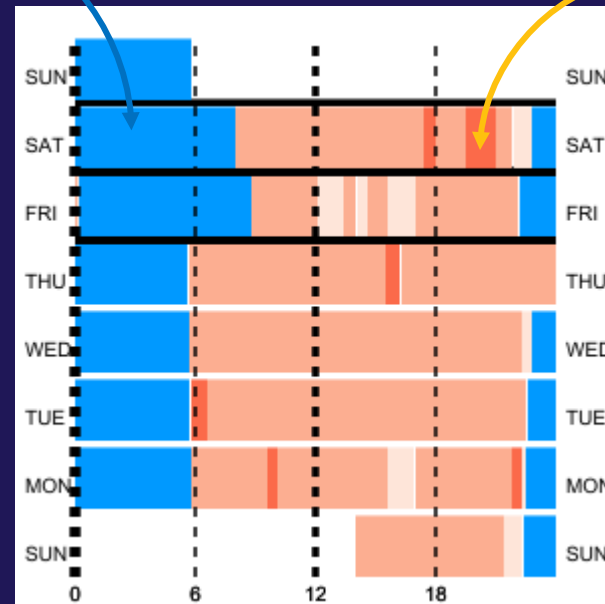


Irregular
sleep and activity

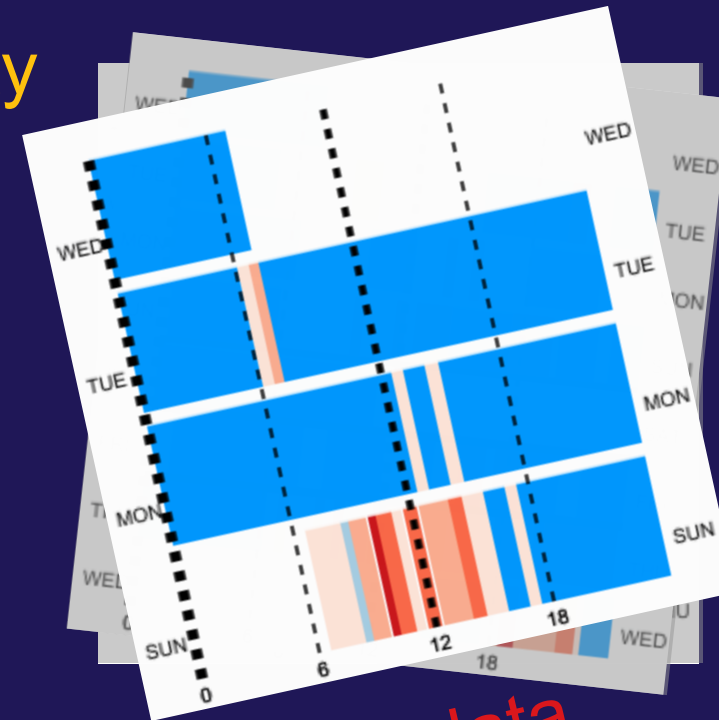
For data-scientists or domain experts

Exploring scientific image data

Sleep



Physical activity

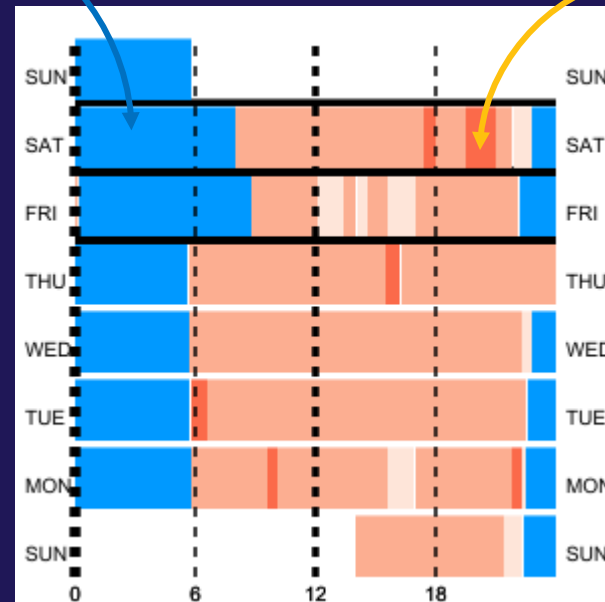


Outlier data

For data-scientists or domain experts

Exploring scientific image data

Sleep

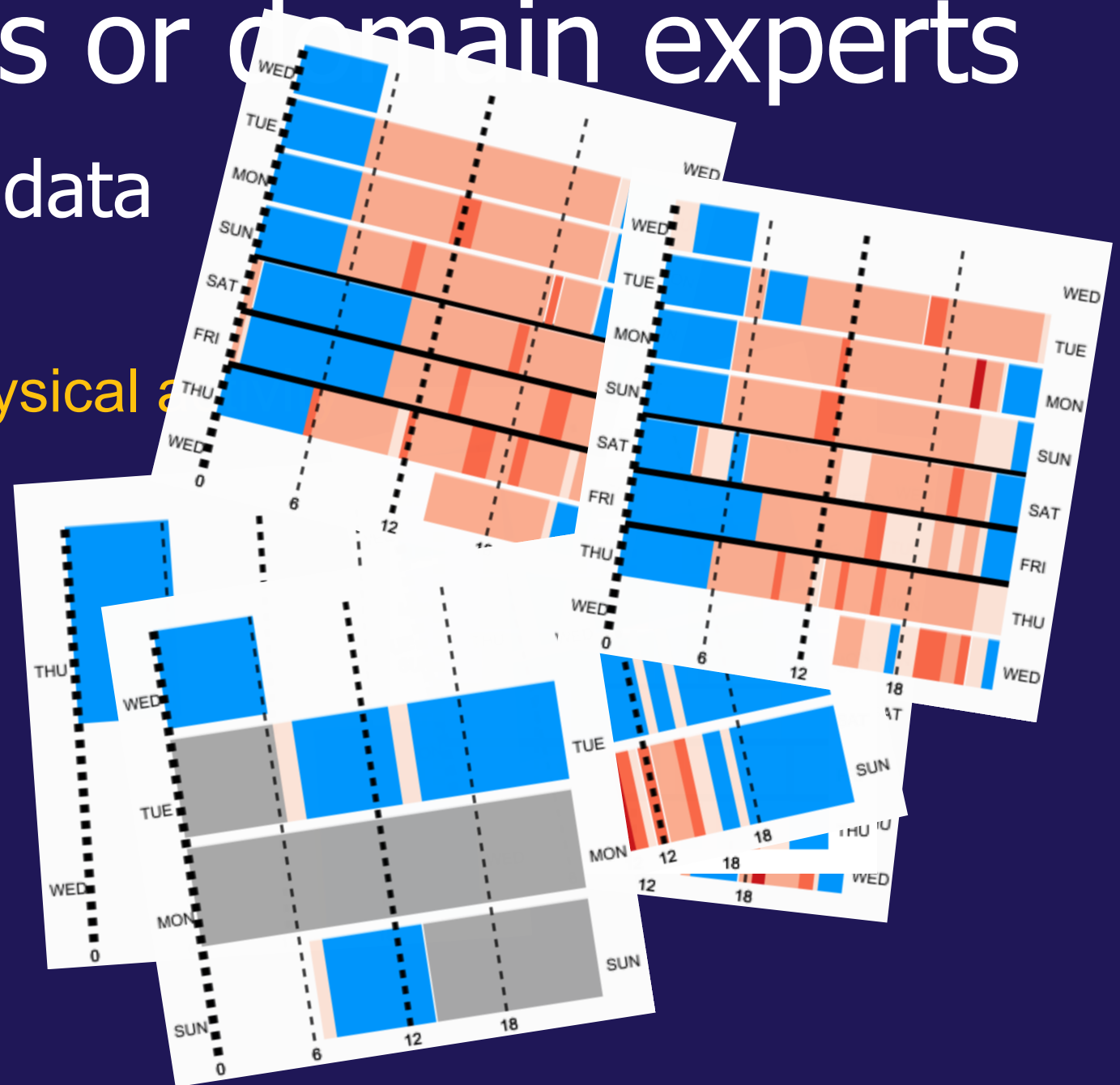


Physical activity



Exploring scientific image data

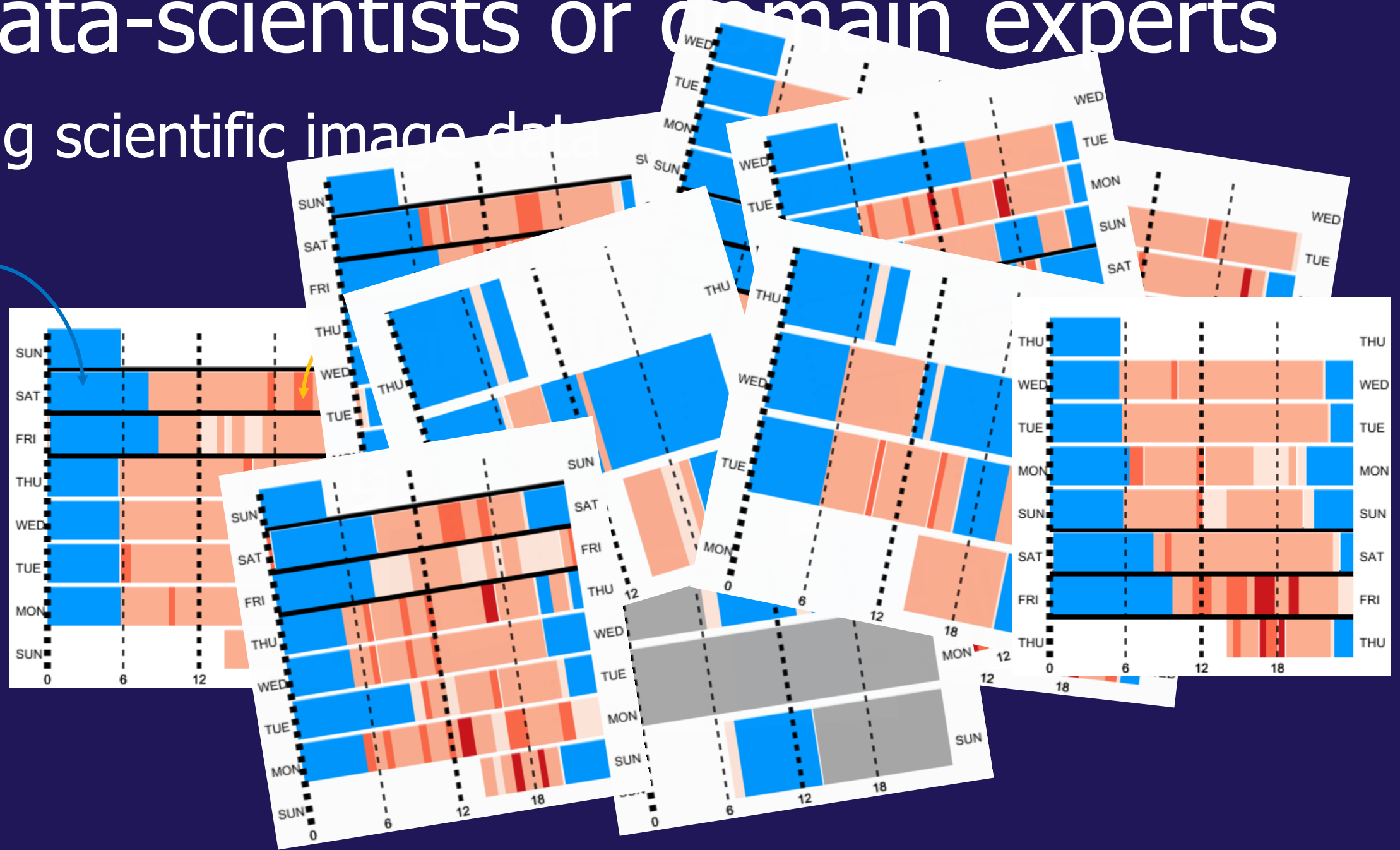
Physical a



For data-scientists or domain experts

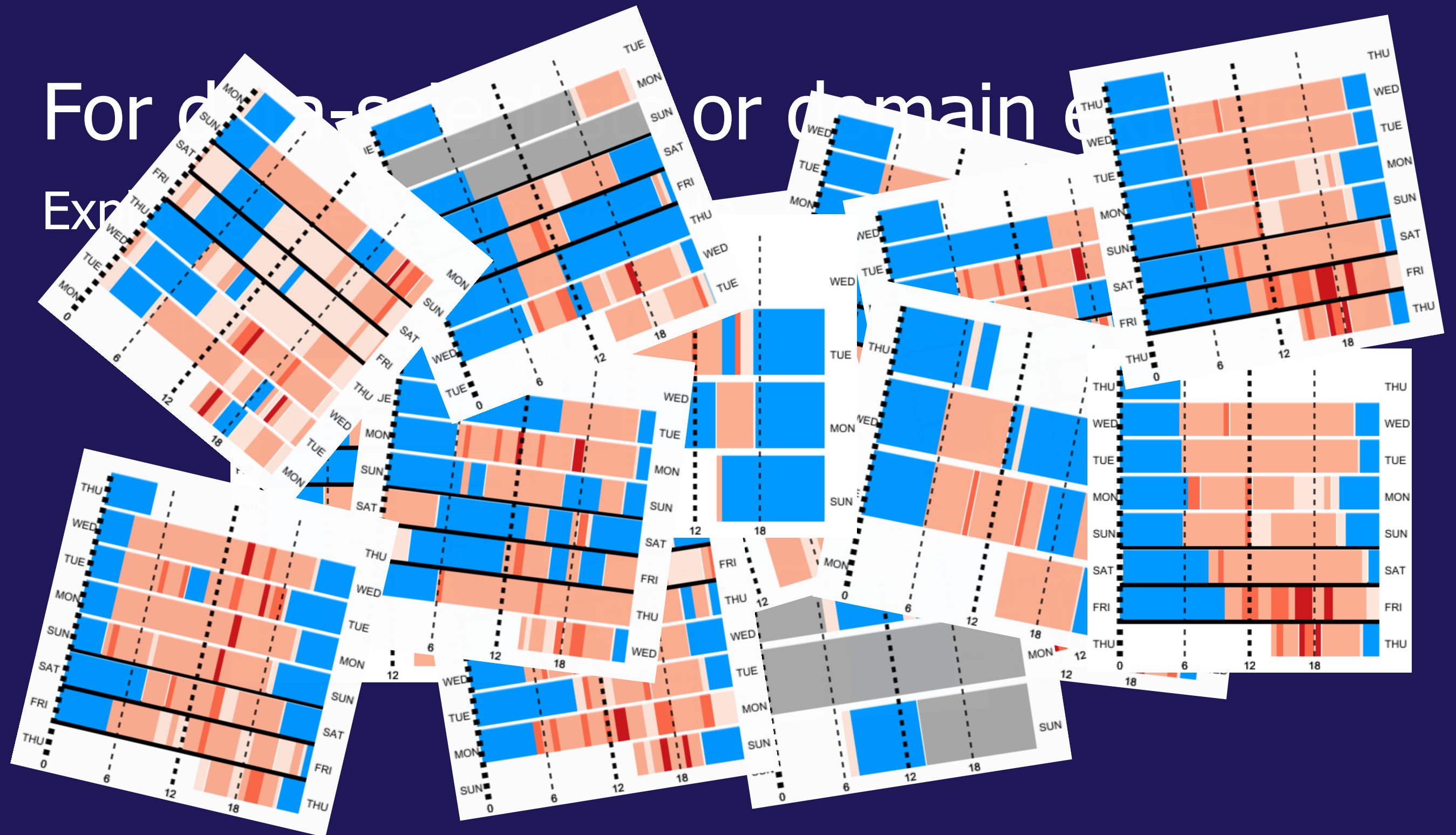
Exploring scientific image data

Sleep



For domain or domain

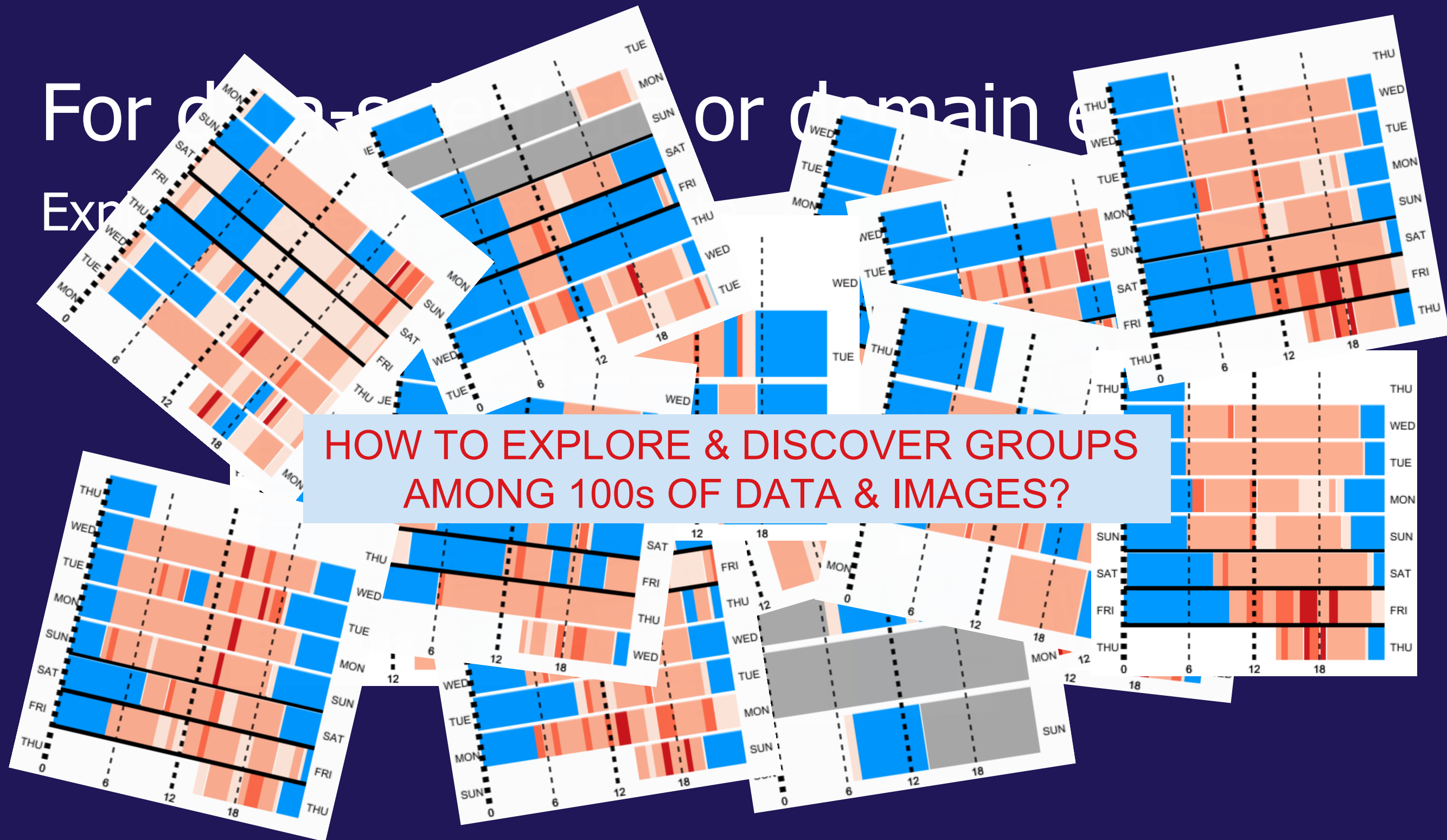
Exp



For data or domain experts

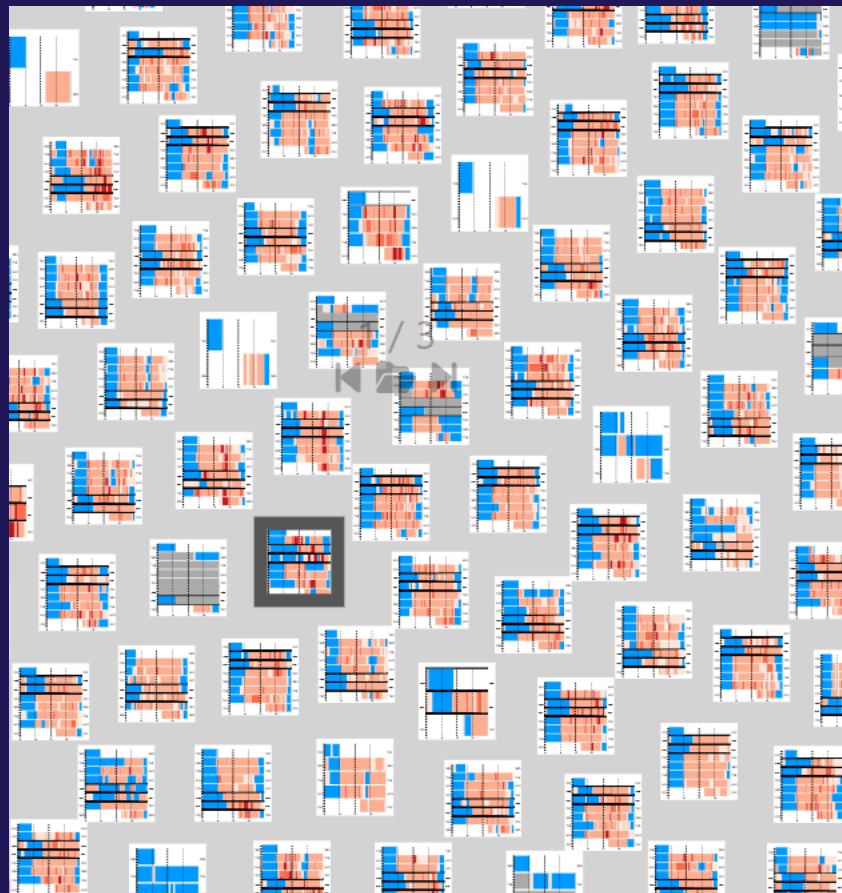
Exp

HOW TO EXPLORE & DISCOVER GROUPS
AMONG 100s OF DATA & IMAGES?



For data-scientists or domain experts

Exploring scientific image data

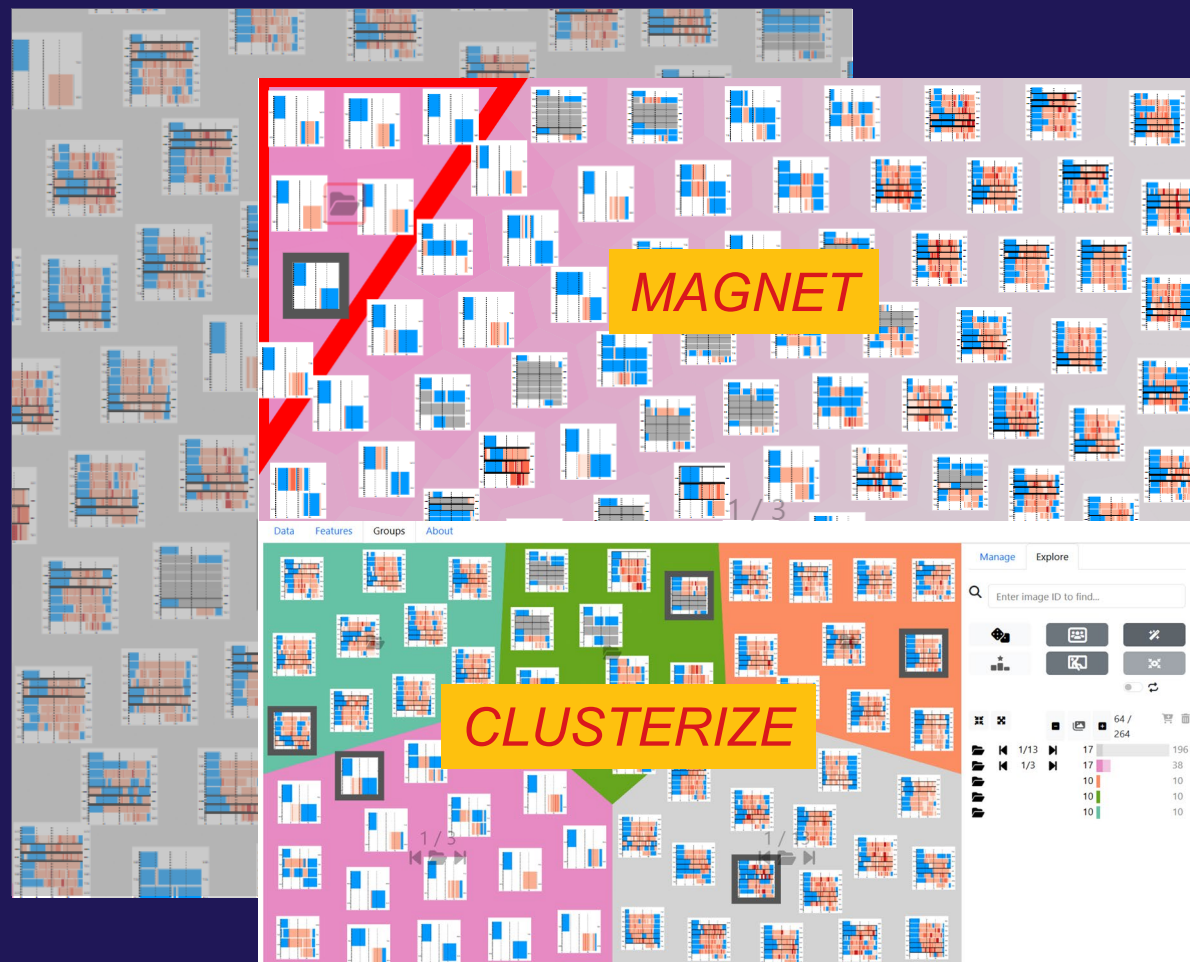


IMAGES & DATA IMPORT

FLUID MANUAL
ARRANGEMENT/GROUPING

For data-scientists or domain experts

Exploring scientific image data



AI-ASSISTED

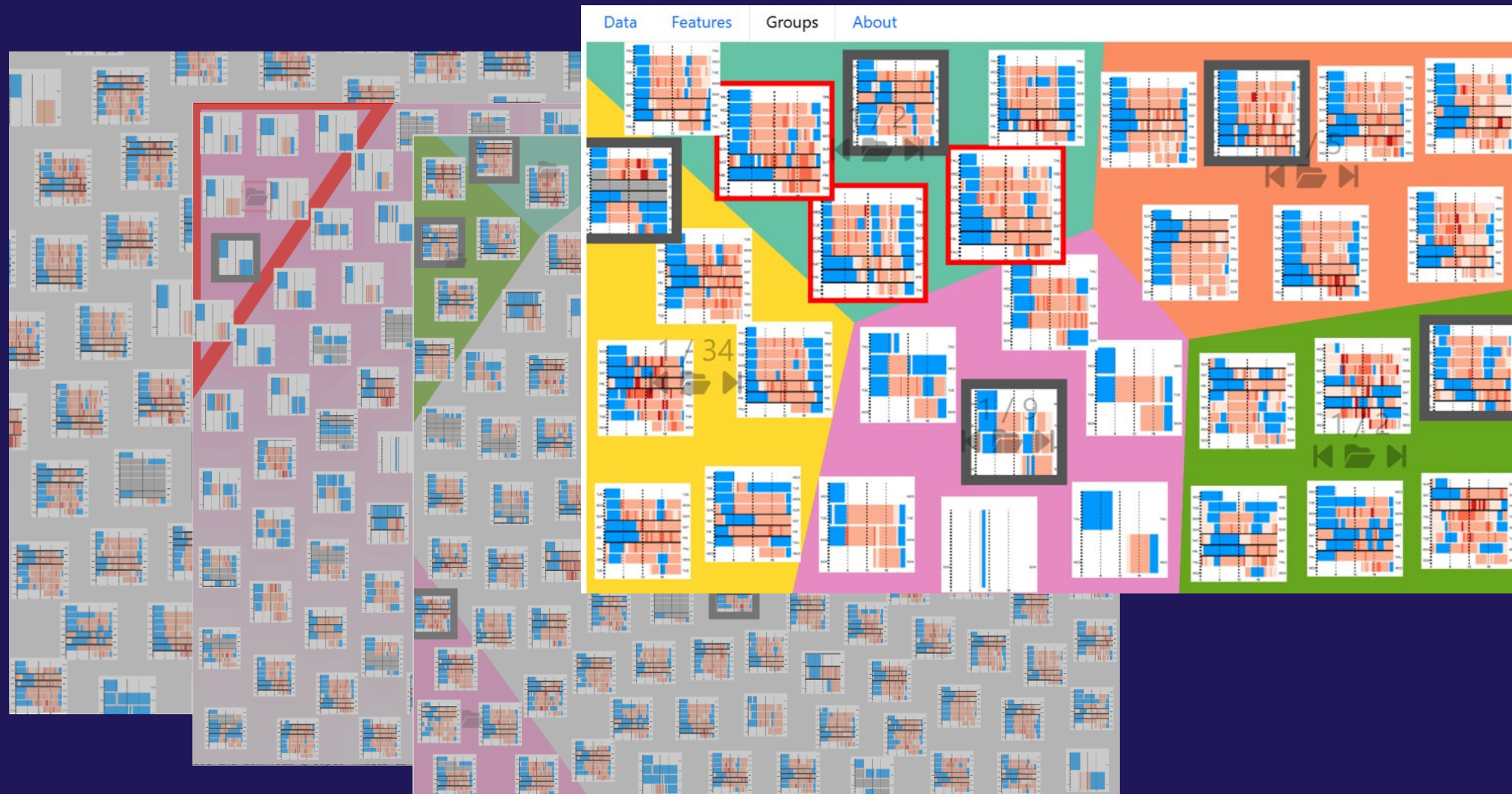
For data-scientists or domain experts

Exploring scientific image data



For data-scientists or domain experts

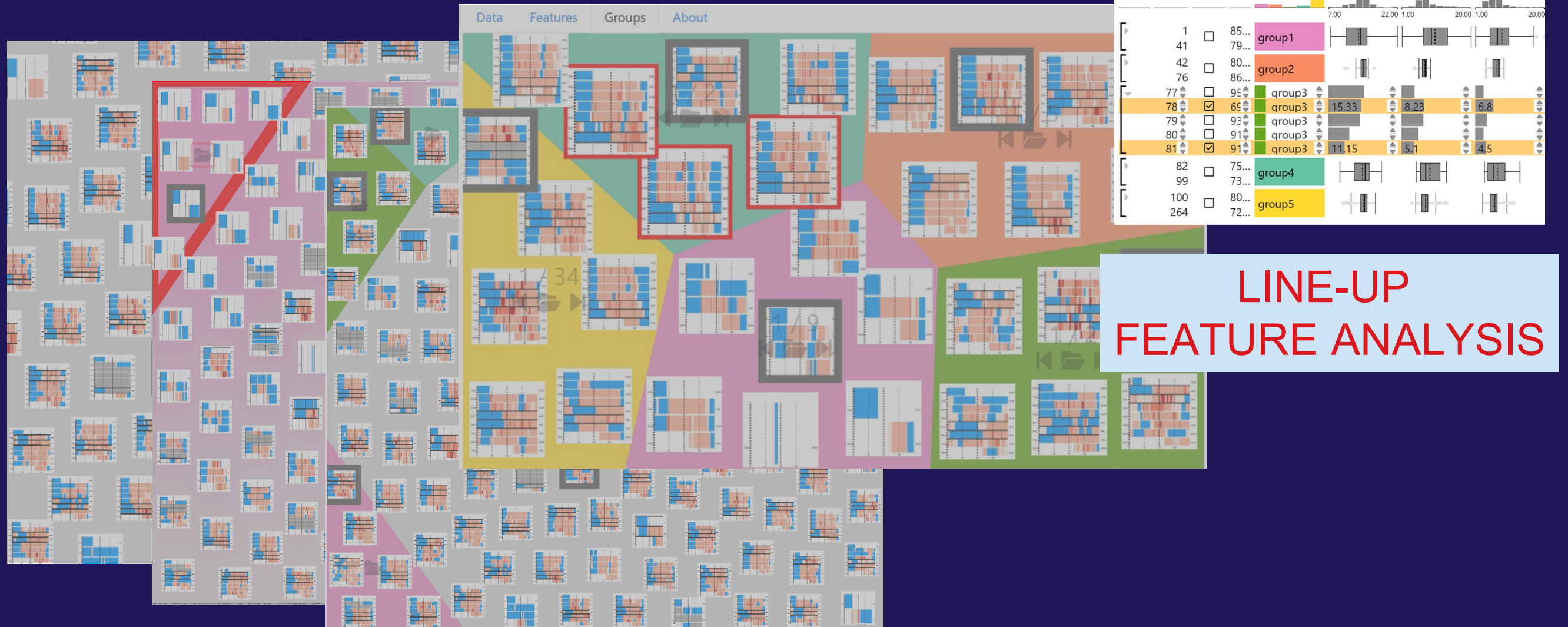
Exploring scientific image data



GROUP
OVERVIEW

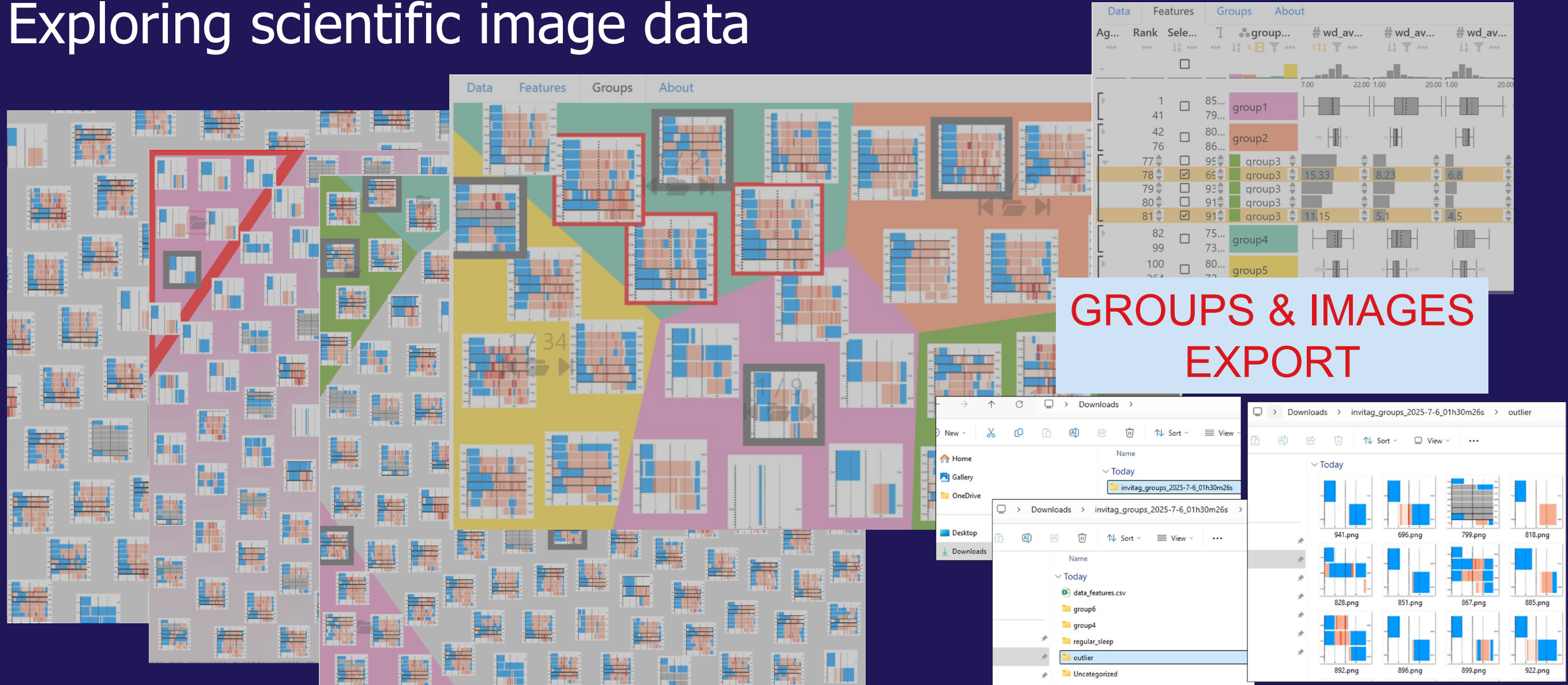
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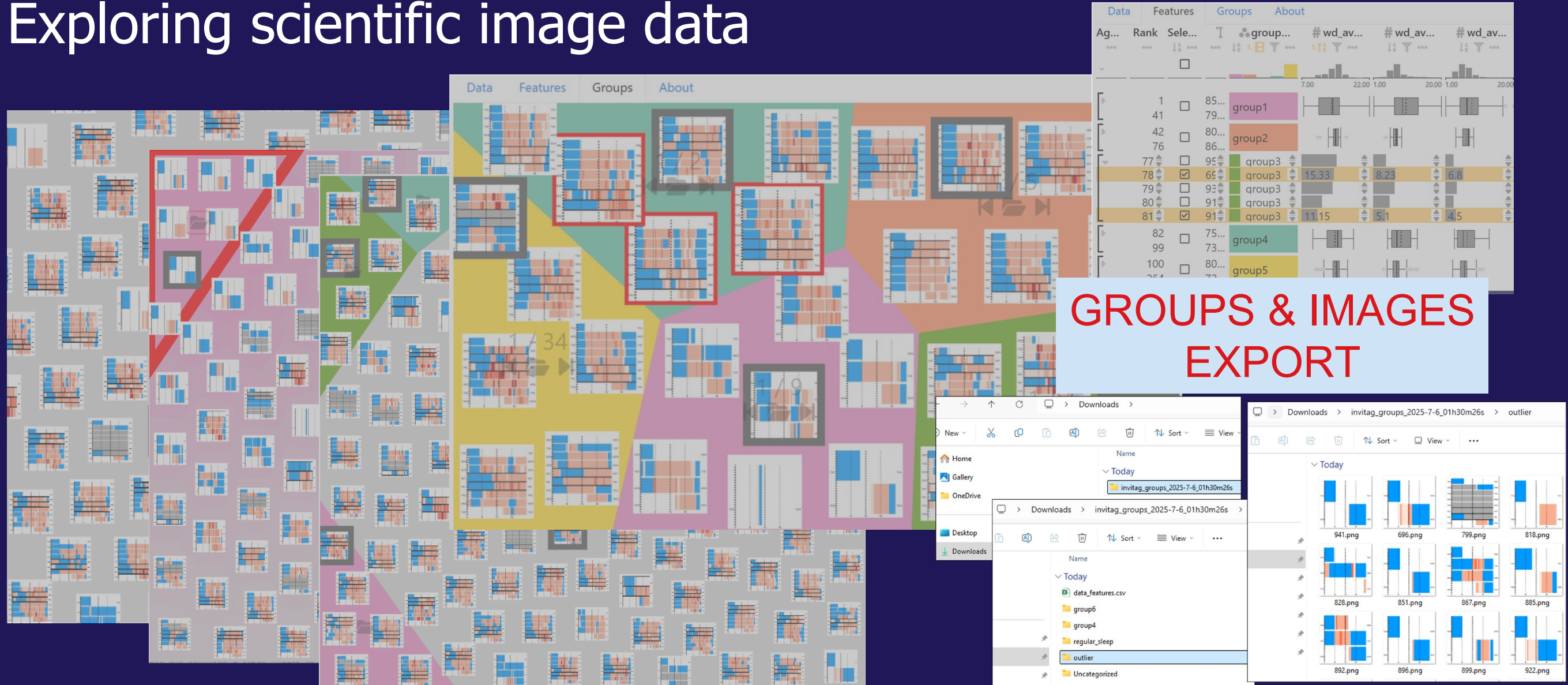
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Data format

User original data
User transformed data

CSV/JSON

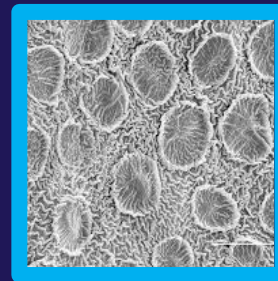
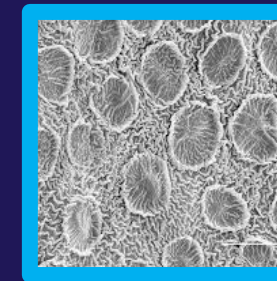
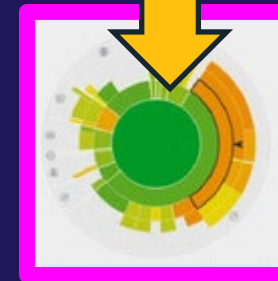
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699	Uncategor #D3D3D3		15.27	8.22	6.8	10	0	21

PNG/JPG/GIF...



(ZIP)

Data + Image embeddings send to our server for AI computing
Data + Image embeddings not stored on our server
Raw images stay on client side
Free to use

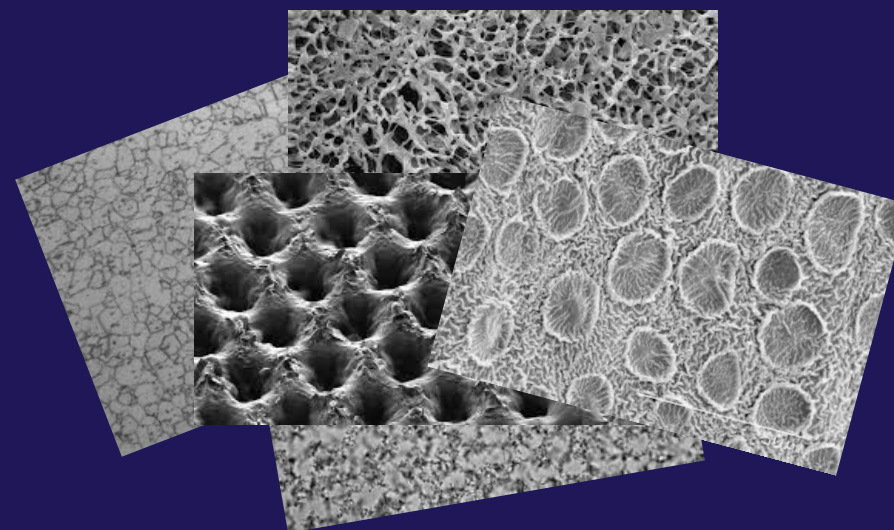
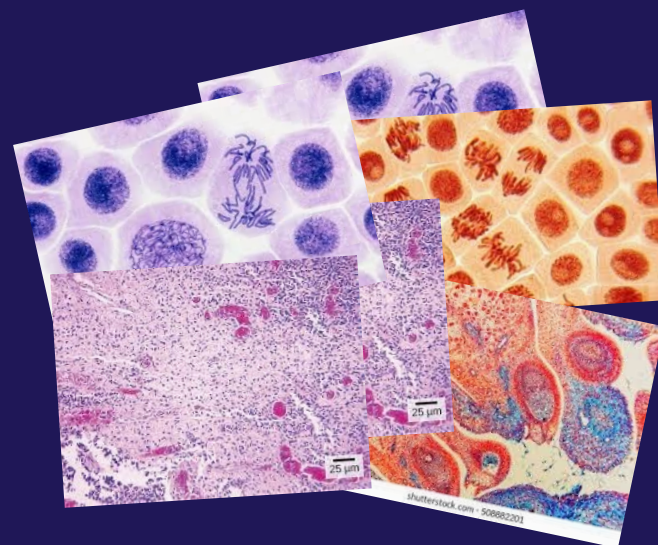
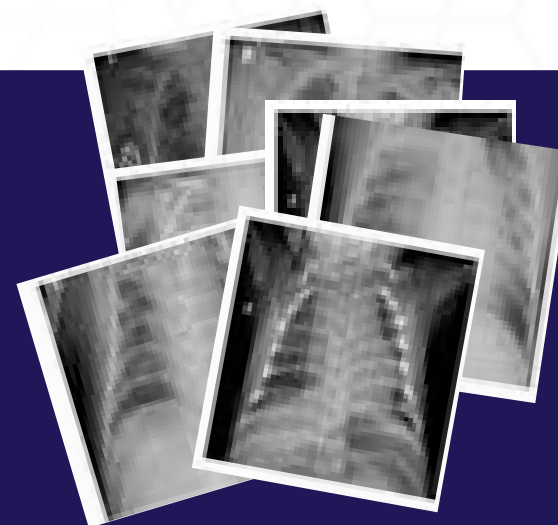
InViTAG

ZIP

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+ Group labels

TRY InViTAG DEMO!



<https://github.com/michaelaupetit/invitag>